

What every figure is actually saying

Three explanations each: **Nephew** (a curious 10-year-old), **Parents** (smart, no background), **Colleagues** (people who know the field).

First, the whole project in three lines

Nephew. When people sort things into piles, they make mistakes. If you look at *which* mistakes they make, you can work out what they were actually seeing in their head. We built a computer that does that working-out instantly instead of taking hours.

Parents. Two people can get the same score on a test and be wrong for completely different reasons — one can't see a feature at all, another sees it but is being too cautious, a third sees both features but can't separate them. There's a mathematical theory that pulls those apart from someone's pattern of errors. Fitting it used to be slow and fragile. We trained a neural network to do it in under a millisecond, and — importantly — to tell you how confident it is.

Colleagues. Amortized neural posterior estimation for General Recognition Theory. A network is trained on simulated 4×4 identification matrices to map directly to the 12 identified parameters with a full posterior, plus an amortized model-comparison head over the 12-node restricted-model hierarchy. Sub-millisecond inference with calibrated uncertainty, which makes GRT usable inside the trial loop rather than only post hoc.

The core suite — [results/figures/](#)

[recovery.png](#)

Nephew. We made up 6,000 pretend people, so we already knew the right answer for each one. Then we asked the computer to guess. Every dot is one guess. If a dot sits on the diagonal line, the computer got it right. Most do — except in the last four boxes, where they're spread out, because those are the hard ones.

Parents. We simulate participants whose true internal parameters we set ourselves, then check whether the method recovers them. Each panel is one parameter: true value across, estimated value up. Points on the diagonal mean a correct recovery. Sensitivities recover almost perfectly; perceptual correlations don't, and that's a limit of the data, not a bug.

Colleagues. Parameter recovery on a held-out showcase set, 200 trials/stimulus, balanced. Twelve panels on fixed ± 3.3 and ± 1 frames with prior bounds marked, so the sign convention (A1/B1 negative, A2/B2 positive) is visible across the grid. Points are hues by the construct at stake in each row — PS(A) for the $\boxed{zx}$ panels, PS(B) for $\boxed{zy}$, correlation structure for ρ — with per-group MAE rather than a pooled r , since correlation is blind to the shrinkage an amortized posterior mean is prone to.

`error_map.png`

Nephew. A grid showing where the computer does well and where it struggles. Dark squares mean big mistakes. The left half is *how big* the mistakes are; the right half is *which direction* they go — too high or too low.

Parents. A heat map: each row is a parameter, each column is a type of participant. Colour shows the size of the error, so you can see at a glance which parameters are hard for which kinds of people. The second half shows whether the errors lean systematically one way, which matters because "wrong by a lot at random" and "always guessing slightly low" are different problems.

Colleagues. Recovery MAE and signed bias by parameter $\times$ generating class, with the two unit families normalised separately (a shared norm makes the ρ block look best when it's worst). Columns ordered by free-parameter count so the complexity gradient reads as a gradient. Cells where $\rho \equiv 0$ by construction are hatched — they measure shrinkage-to-zero, not recovery. The bias panel is there because MAE cannot distinguish systematic shrinkage from unbiased noise.

`confusion.png` and `confusion_constructs.png`

Nephew. There are 12 different kinds of pretend people. The computer has to guess which kind it's looking at. This grid shows how often it guesses right, and when it's wrong, which kind it mixes it up with. The second picture breaks it into three easier questions instead of one hard one.

Parents. The 12 participant types differ along three yes/no-ish dimensions. The first figure asks "did it name the exact right type?", which is harsh, because some types are genuinely almost indistinguishable from their neighbours. The second asks the three underlying questions separately, and that's the fairer picture: it gets two of them right about 95% of the time and struggles with the third.

Colleagues. Row-normalised 12-way identification, complexity-ordered so nested neighbours are adjacent and off-diagonal mass falls in a readable band. Classes never selected are daggered to distinguish a structurally empty column from one that's simply never confused. Exact 12-way accuracy is demoted to the subtitle beside the three per-construct accuracies, since exact-class accuracy penalises distinctions between statistically indistinguishable nested models. The companion figure gives the $2 \times 2 / 2 \times 2 / 3 \times 3$ construct confusions directly.

`calibration.png`

Nephew. The computer doesn't just guess — it says how sure it is. This checks whether it's telling the truth about that. When it says "I'm 90% sure", it should be right 90 times out of 100. Not 60. Not 99.

Parents. A confident wrong answer is worse than an uncertain one, so we test the honesty of the uncertainty. If the method says the truth is inside a range 90% of the time, we check how often it actually is. On the graph, being on the diagonal means honest. The sensitivities are honest; the correlations are slightly overconfident.

Colleagues. Simulation-based calibration. Rank histograms per parameter family with a Binomial 95% band, so uniformity is assessed against a tolerance rather than by eye, plus nominal-vs-empirical interval coverage per family. Families are separated because pooling lets conservative z ranks cancel overconfident ρ ranks. ρ values that are exactly zero under PI are excluded — the rank there measures posterior mass below zero, not calibration.

`uncertainty.png`

Nephew. The more questions you ask someone, the more sure you can be about them. This shows the computer getting more confident as it gets more data — and where it stops improving no matter how much you give it.

Parents. More trials should mean tighter estimates. This plots how much the uncertainty shrinks as data accumulate. The dotted guide line is the rate you'd expect from statistics alone; where the real line flattens away from it, we've hit the floor of what the network itself can resolve.

Colleagues. Posterior SD vs total trials on log-log, per family, median with IQR band. The $n^{(-1/2)}$ reference anchored at the first bin makes departure from root-n scaling — the amortization floor — legible as a departure from a straight line. Quantile bins merged forward to a minimum count, with per-bin n printed.

`constructs.png`

Nephew. Two pictures. The left one checks whether the computer's "how sure am I" numbers are honest. The right one shows the thing it's genuinely bad at: telling whether two features are seen separately or get tangled together. When they're barely tangled, it can't tell — and neither could anyone.

Parents. Left: are the reported probabilities trustworthy? Right: the honest limitation. Judging whether two perceptual features are independent gets easier the more strongly they're actually related, and near-independence is close to unknowable from this kind of data. We report a probability instead of forcing a verdict, precisely because of this.

Colleagues. Reliability curves for the three construct probabilities with Wilson 95% intervals and marker area scaling with bin count, plus the PI identifiability frontier in fixed 0.05-wide $|\rho|$ bins. True-PI cases are drawn as a separate marker rather than joined to the curve, since $\rho \equiv 0$ is a distinct condition and not a point on a magnitude axis. Ground truth comes from `MODEL_SPECS`, and the decision threshold is stated on the panel.

[speed_accuracy.png](#)

Nephew. How long the computer takes versus how long the old way takes. The bars go up by factors of ten, so a bar twice as tall is a hundred times slower.

Parents. The old method fits every candidate model and picks a winner, which takes around half a second per participant. Ours takes under a millisecond. Two bars for ours because timing one at a time and timing a thousand at once give different answers, and the one-at-a-time number is the honest one for real-time use.

Colleagues. Wall-clock per matrix, log scale, against both MLE workflows — the saturated [fit_full](#) and the realistic AIC/BIC-selected fit. GRIN reported batched and single-matrix, because batched throughput divided by batch size is not a latency claim.

[results/figures/generation/](#)

[coverage_report.png](#) and its seven panels

Nephew. Before we test the computer, we check we gave it a fair variety of pretend people — not all easy ones, not all the same.

Parents. Diagnostics on the simulated training set: are the participants we invented spread across the whole range of possibilities, or clustered somewhere convenient? Each panel is one property, with the edges of the allowed range marked so you can see whether we're using the full space.

Colleagues. Prior predictive coverage. Marginals over accuracy, per-stimulus accuracy, congruency asymmetry, response bias, $|z|$, ρ , and trials per stimulus, with [Z_MAX](#) and [R_MAX](#) drawn so the prior's support is visible rather than inferred.

[results/figures/recovery/](#) — how we compare to existing software

[recovery_grin.png](#), [recovery_mdstd.png](#), [recovery_grtools.png](#),

[recovery_python_mle.png](#)

Nephew. The same test, given to four different computer programs, on exactly the same pretend people. Now you can put the pictures side by side and see who did best.

Parents. Two established R packages that researchers currently use, plus a straightforward maximum-likelihood implementation, plus ours — all run on identical data so the comparison is fair. The marker shape shows whether that method also identified the right participant type, so you can see estimation quality and model-choice quality at once.

Colleagues. Per-method recovery grids on identical stratified matrices with matched geometry and axes. Hue by construct, shape by per-construct classification correctness.

Headline statistics computed on the subset where every method converged, since scoring each method on its own survivors systematically flatters whichever fails most.

[summary_recovery.png](#) / [summary_classification.png](#)

Nephew. The scoreboard. Who won, and where.

Parents. Aggregate comparisons — accuracy by parameter type, by how much data was available, and by how strong the true correlation was. The classification one shows not just how often the methods *agree* but how often they were both actually *right*, which is the question that matters.

Colleagues. MAE by family, trial bin, and $|\rho|$ bin with bootstrap CIs; per-construct and 12-way accuracy; and the agreement decomposition (both correct / reference only / baseline only / both wrong). Bare agreement is uninformative where ground truth exists.

[results/figures/rt/](#) — adding response times

[rt_recovery.png](#), [rt_confusion.png](#), [rt_calibration.png](#), [rt_uncertainty.png](#)

Nephew. The same tests as before, but now the computer also gets to know *how long* someone took to answer, not just what they answered.

Parents. Identical diagnostics to the main suite, run on a version of the model that also sees reaction times. Directly comparable to the counts-only figures.

Colleagues. Parity suite for the RT-augmented model on held-out matrices, same evaluation protocol.

[rt_vs_counts_error_gain.png](#)

Nephew. A map of where knowing the timing helped and where it didn't. Warm squares mean it helped.

Parents. The same participants analysed twice — once with just their answers, once with answers plus timing — so any difference is due to the timing alone. Colour shows where the extra information paid off.

Colleagues. Diverging map of MAE reduction by parameter $\times$ class on paired matrices, unit families normalised separately, symmetric about zero so "no change" reads as white.

[rt_vs_counts_paired.png](#)

Nephew. Timing helped on average — but did it help *everyone*, or help a few people a lot and make others worse? This shows that.

Parents. Averages hide things. A small average improvement is consistent with "helps a bit for everyone" and with "helps enormously for a third and hurts the rest". These are different claims and only the distribution tells them apart.

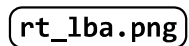
Colleagues. Per-matrix paired gain distributions with median and the proportion improved, which is the honest headline for a paired comparison.



Nephew. With timing, the computer can also work out *how* someone is looking — whether they check both features at once or one after the other. Answers alone can't tell you that at all.

Parents. Reaction times carry information about processing strategy that accuracy alone simply doesn't contain. This shows how well that strategy is recovered, against chance.

Colleagues. Five-way SFT architecture confusion with balanced accuracy against chance, plus the counts-vs-RT gain split by unit — accuracies with chance levels marked, ρ as MAE on its own axis.

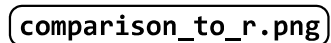


Nephew. Checking the computer also got the timing-related numbers right — how fast someone gathers evidence, how much they need before deciding.

Parents. The reaction-time model has its own parameters (evidence accumulation rate, decision threshold, and so on). Same recovery check as before.

Colleagues. LBA parameter recovery, r and MAE per parameter, matched square frames.

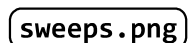
The two standalone comparisons



Nephew. Four scoreboards against the programs researchers use now: how fast, how often it breaks, how accurate, and how often it agrees *and* is right.

Parents. Speed, reliability, accuracy, and agreement. The reliability panel matters more than people expect — the existing tools sometimes fail to produce any answer at all, and ours always returns something with an honest uncertainty attached.

Colleagues. Speed (batched and serial), convergence failure rate with Wilson intervals, MAE by trial regime on the common-convergence subset, and correctness-decomposed model selection agreement.



Nephew. Checking we didn't just get lucky — we changed the settings lots of different ways and it kept working.

Parents. Robustness checks. Does the result survive changing how much data, how wide the assumed range of participants, how big the network, and which random starting point? If it only worked for one specific setup, that would be a problem.

Colleagues. Sweeps over trial count, prior envelope, network capacity (shown as seed spread), and RT speed regime.

Poster figures — presentations/sbi_poster/figures/

[poster_recovery.png](#)

Nephew. Same as the recovery picture, squashed down so it's readable from across a room.

Parents. A condensed version: all the sensitivity parameters pooled into one panel, all the correlations into another, so the two numbers on the poster are exactly the two numbers in the figure.

Colleagues. Family-pooled recovery so the printed r matches the quoted r , on the same fixed frames as the manuscript version.

[calibration.png](#) (**poster version**)

Nephew. Just the honesty check, nothing else, big enough to read.

Parents. The coverage half of the calibration figure. On the diagonal means honest.

Colleagues. Coverage-only variant; the rank histograms are the right thing for a methods section and the wrong thing for a three-second read.

[accuracy_crossover.png](#)

Nephew. How wrong each method is as you give it more and more data. Lower is better. Ours stays lower the whole way.

Parents. As data accumulates, both methods improve. The gap narrows but doesn't close over the range we tested. The advantage comes from the method having sensible expectations built in, which matters most when data are scarce.

Colleagues. MAE vs trials/stimulus, log-log. The Bayes estimator's advantage over ML under a correctly-specified prior, with the ratio contracting as n grows. See the note in the accompanying analysis about the MLE optimiser's bound-hitting rate before quoting the magnitude of the gap.